

A Conceptual Framework for Test-Taker Experience in Educational Testing

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Abstract

Practitioners in educational testing are giving more attention to the experiences of test-takers as users of tests, following the practices in Human-Computer Interface (HCI) and User Experience Research (UXR) applied to educational measurement (Burnstein et al., 2021; Cheng & DeLuca, 2011; Mislevy, 2018). Concepts like test-taker experience (TTX) or assessment experience are now popular, but little research is available explaining what exactly those terms mean. In this paper we present a conceptual framework for test-taker experience in educational testing. Our model includes three dimensions: temporal stages, types of interactions, and levels of sense-making. We applied this framework to analyze test-takers' experiences shared on Twitter about a test used in Chile for college admission. Results show at least four types of interactions were represented in the tweets with acceptable levels of congruence among raters. We conclude with a more general discussion about the value of studying test-taker experiences in educational measurement.

1 Introduction

Educational Technology has evolved in vertiginous ways in recent decades, extending its reach to almost all facets of Education. The influence of Human-Computer Interface (HCI) into Education has been accompanied by the extension of the use of educational technologies. Certain phenomenological ethos, concerned with how users actually use and live within the new technologies we generate, have continued a more general project for education in which student experience is at the center (Dewey, 1938). Curriculum, Instruction, and Evaluation have also been carried into this post-phenomenological turn, in which technologies are better understood through the experiences of their users (Idhe, 2009).

The general idea of studying interactions in educational technology —between students and technologies or among students in the context of use of certain technology for learning— is quite clear in terms of its relevancy (Borokhovski et al., 2016; Wu et al., 2019; Lo et al., 2024; Wei et al., 2024). However, this ethos towards empathy and understanding of inner experiences on the user side has not necessarily been as quick for Educational Testing and Assessment as with other sub-fields in Education.

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As many tests moved to computer-based formats (Terzis et al., 2012), it was inevitable at some point to bring this “user-centered”, “interaction-centered” view into educational testing. Many criticisms have been made against standardized testing (Cronbach, 1975; Linn, 2001), including a general perception that tests (and particularly, high-stakes standardized testing) create bad user experiences. Concerns about anxiety (Anis et al., 2016; Fulton, 2016; Hilliard et al., 2020; Pate et al., 2021), stereotype threat (Good et al., 2003; Steele & Aronson, 1995; Stricker & Ward, 2004; Wasserberg, 2017), or test-taker construction of identity towards tests (Holmes, 2021; Huhta et al., 2006), among many others, are all topics that suggest the need of empathizing better with test-takers.

Practitioners in Game-Based Assessment were pioneers in this ethos of empathizing with test-takers as they created technologies for assessment. Even if initial efforts focused mainly on the evidentiary part of assessments (how valid are the interpretations and uses; Mislevy et al., 2003; Mislevy et al., 2016), the practice of Game-Based Assessment recurred to the practices found in game design (DiCerbo, 2014; Ifenthaler et al., 2012; Kim & Shute, 2015; Ventura & Shute, 2013).

In some way, we can trace this trend towards empathy with test-takers in the test engagement literature (e.g., Wise & Kong, 2005; Wise, 2017; Wise & Kuhfeld, 2021), focusing on the validity problems of not empathizing and quantifying disengagement. Mislevy’s (2018) socio-cognitive foundations of educational measurement also brings a similar spirit towards empathy, elucidating how the cognitive and the cultural are related. The inclusion of test-taker voices into validation practice and test-taker perception studies (Araneda, 2023; Cheng & DeLuca, 2011; Gardiner & Howlett, 2016; Song, 2018; Xie, 2011) can be also considered part of this shift, trying to connect with test-takers in a more systematic way.

The concept of Test-Taker eXperience (TTX) proposed by researchers at Duolingo (Burstein et al., 2021) crystalized this trend, elevating the matter of user-experience to all parts of test design, and importing this into Language Assessment. A similar acknowledgement can be found in members of Educational Testing Service, advocating for a “user-obsessed” paradigm in the design of assessment technologies (Marsh Runyon et al., 2022; McWilliams, 2020). Related to this as well is the concept of “caring assessment” (Zapata-Rivera, 2017).

However, with the increasing use of experience as a conceptual category, there has been little discussion on what we mean by the word “experience.” To define experience is a challenge on its own, in which philosophers have failed to come up with a strong consensus. Despite those failures, there is value in taking some time to understand what we mean with the word experience as in “assessment experience” or “test-taker experience”, what it encompasses, and — if possible — define a conceptual framework to refer to it and have a common vocabulary. Other fields have tried this path, and now is the turn of educational testing to build a conceptual framework to include the phenomenology of being assessed in test development and validation.

In this paper, we propose a conceptual framework for test-taker experiences in educational testing. The proposed conceptual framework is based on other frameworks found in the literature conceptualizing experience in their own context. We took those frameworks and merged them into the one we propose here, adapted to the needs of educational testing.

In addition to methods found in UX research, we also borrow from Dewey’s three main principles of experience (Dewey, 1938), to be explained later in this paper. We proceed by first reviewing the literature on conceptual frameworks for UX. We then propose a model for using UX methods in test development. Next, we explain how the methods we use in educational assessment can be mapped into our framework. We provide some empirical work using the framework identifying UX experience of test-takers in Chile, and statistical work analyzing the consistency of two raters using the conceptual framework we propose. We end with a discussion on how validation can take advantage of the rich evidence coming from UX research.

1.1 Theoretical Basis for the Concept of Experience

The study of test-taker experiences requires familiarity with some philosophical concepts. The definition of “experience” can relate to two different concepts: (a) the more immediate and instantaneous stream of information a person can perceive; or (b) a narration about something that happened, which represents an integration of the raw stream of information into a more cohesive and curated narrative of what is happening or just happened. Jay (2005) noted this difference in the etymology of the word *experience* in German. *Erlebnis* and *Erfahrung* both translate into English as experience but correspond to quite different notions of experience. *Erlebnis* is more related to “lived experience,” with a sense of unity and lack of differentiation or objectification. It generally connotes a more immediate, pre-reflexive, and personal variant of *Erfahrung*. *Erfahrung* corresponds to some cognitive judgment about the “lived experience.” It relates to some learning process, integrating discrete moments of experience into a narrative whole or an adventure. Here, there is some role of memory in the perception of the experience. *Erfahrung* is related to a “dialectical notion” of experience (Jay, 2005, p. 11).

As we describe later, both concepts of experience have relevance to test development and validation. Several authors who proposed putting users’ experience at the center of a design process borrow the concepts of experience from Dewey. In his book *Experience and Education*, Dewey (1938) proposed a new type of education that acknowledged the connections between education and personal experience. In his philosophy, experience has two main principles: a principle of continuity and a principle of interaction. The principle of continuity refers to experience lived as a continuum with past experiences and expectations of future experiences. It assumes all our instantaneous experiences, both past and future, are part of the conscious process by which we process the raw stream of information we experience.

Second, the principle of interaction says the experience happens as we interact with our environment, and it assumes there is no experience without interaction. Interaction refers to the cerebral or corporal activity that happens in the specific encounter of a person and a situation. Both concepts are part of Dewey’s ideas for education in that every educational experience needs to be done in a way that makes sense with past experiences, allows for growth, and does not block future educational experiences. Dewey considered the difference between a generic experience and “an experience” as a moment with a beginning and end.

Forlizzi and Battarbee (2004) proposed a theoretical framework for designing experience in interactive systems, inspired by Dewey’s work. They defined an interaction-centered model in contrast to user-centered approaches found in the user experience literature. The difference is that, to characterize and cast an experience, a designer considers users as isolated versus users who will interact with products. They defined three types of interactions between users and products: *fluent*, *cognitive*, and *expressive*. Fluent interactions refer to a user’s well-learned interaction, unconsciously done when facing a product (e.g., riding a bicycle). A cognitive interaction occurs when the user’s cognitive resources are needed when interacting with a product (e.g., when trying to understand the functioning of a new device). Expressive interactions help the user form a relationship with a product, such as personalization in the presentation of a device or avatars. As described later, considering a testing situation as an interaction between a user and a test helps to study test-taker experiences.

1.2 Co-construction and Sense-making of Experiences

Forlizzi and Battarbee (2004) described three types of experience: *experience*, “an experience,” and *co-experience*. Experience and “an experience” correspond to the two definitions of experi-

ence mentioned earlier in German etymology. Co-experience is a distinct idea that considers how experience is processed and evaluated collectively with others. As they explained,

Social situations greatly influence co-experience. For example, whether running out of gas when driving to the countryside with friends is viewed as an adventure or a disaster depends on how the friends decide to interpret the situation. One person might be upset, another might point out the humorous potential, and a third might agree. (p. 263)

In the context of educational assessment, this co-construction of experience can occur among several actors involved in a testing process (e.g., peers, teachers, parents, or school administrators).

Wright and McCarthy (2010) also emphasized co-experience in product design. They proposed a design paradigm called *experience-centered design*, in which the designer and user engage in a dialogical process to create an artifact. They also discussed how a user’s experience is shaped by this dialogue—explicit or implicit—between designer and user. Because any person can become a co-creator of experience, there is the possibility of forming a co-experience with the designer of a product. We refer to this phenomenon as the *dialogical experience*.

A third principle, not explicitly stated by Dewey but inferred by subsequent authors as part of his conceptualization of experience, is that individuals continually make sense of their experiences. While experiencing something, people engage in an underlying meaning-making process that integrates incoming information, ongoing interactions, connections to past experiences, and expectations of future ones. The notions of “an experience” and co-experience in Forlizzi and Battarbee’s (2004) framework, as well as the dialogical experience described by Wright and McCarthy (2010), all incorporate this principle that humans actively interpret and make sense of their experiences.

1.3 A Conceptual Framework for Test-taker Experience

This section draws from previous conceptions about experience to propose a conceptual framework for test-taker experience. The proposed conceptual framework is a starting point for understanding examinee’s experiences in educational testing, that most likely will need further refinements.

In our proposed framework, we draw from the previous works cited to define different levels in the sense-making process. We specify five levels of sense-making:

- **Free-stream** refers to the rawest and instantaneous experience, a stream of self-talk. Think-aloud protocols are an example of a method that taps in this level, barely made sense.
- **Immediate** refers to the experience that just happened, on which there will be some level of evaluation, of integration of the raw experience, but is still incomplete. Test-takers’ report right after they respond to an item corresponds to the Immediate level of sense-making.
- **Integrative** refers to the dialectic experience without co-construction of the experience. It refers to a more constructed narrative of an experience with a beginning and end, done in isolation from other stakeholders.
- **Co-integrative** refers to a dialectic experience enriched with the co-construction with others. This level appears in focus groups or some modes in ethnographic research.
- **Dialogical** corresponds to the most structured and long-term evaluation of experience that includes the experience affected by implicit and explicit dialogues with the test designer.

Table 1 summarizes these levels of experience, ordering them in terms of the increasing complexity of construction. Table 1 also illustrates that the first two levels can be studied at the item or activity level, whereas the three higher levels refer to the overall test-taker experience.

Table 1: Levels of Sense-making in the Test-taker Experience

Level of Experience	Description	Investigative Focus
Free Stream	Self-talk, the continuous stream of information and responses to the stimuli.	Item or Activity
Immediate	An initial and incomplete integration of a short experience that just happened.	Item or Activity
Integrative	A more complete and integrative evaluation of an experience that has a clear beginning and end, expressed as a story.	Overall Testing Experience
Co-integrative	Similar to integrative, but now the story is built including the voices of other persons, integrating them in the process of making sense of the free-stream, unitary experience.	Overall Testing Experience
Dialogical	Similar to co-integrative, but now the other is the test designer in particular, both making sense of the experience in an empathetic dialogue.	Overall Testing Experience

In addition, we define five types of interaction between an examinee and a test:

- **Fluent interactions** happen in an automatic or pre-conscious way (e.g., riding a bicycle is an interaction between user and “product” that is essentially automatic after one learns how to ride a bike).
- **Cognitive interaction** involves a cognitive process triggered by the encounter with the object. This is the primary type of interaction expected with the test in educational testing.
- **Emotional interaction** refers to emotions in the specific encounter between an individual and a test. Anxiety, disengagement, boredom, and fun correspond to an emotional interaction.
- **Expressive interactions** are manifestations of self-expression in reaction to the test. An opinion appearing in the mind of an examinee in response to a test item is an example of expressive interaction.
- **Physical interaction** corresponds to interactions that happen within the body. In the context of a test, having the need to go to the bathroom, feeling hungry, cold, or hot are examples. In some cases, a test may demand physical actions by the examinee as its primary expected interaction (e.g., physical abilities tests for firefighters, police, or the military).

These five types of interactions borrow the fluent, cognitive, and expressive interactions proposed by Forlizzi and Battarbee (2004) and the emotional interaction from what Holbrook and Hirshman (1982) name ‘affect’ in their experiential approach to marketing. In our framework, we add physical interactions to account for physical situations occurring during an exam, such as gut pain due to anxiety or exams involving physical responses.

The continuum of past and future experiences involves characterizing and studying the relationship between past experiences, expectations of future experiences, and the instantaneous experiences that occur while taking a test in its many “stages.” There are six stages relevant to the test-taker experience:

- **Ante-experience** refers to experiences related to a test well before an examinee starts to prepare for it.
- **Pre-experience** refers to experiences preparing for the test; that is, experiences when an examinee is sure he or she will take the test.
- **Zero-experience** refers to the experience while taking the test.
- **Inter-post experience** refers to the experience between taking the test and receiving the results.
- **Post experience** refers to the experience right after receiving the test results.
- **Long-term experience** refers to the experience after a reasonable amount of time since receiving the results of a test.

The levels of sense-making, types of interactions, and temporal stages of test-taker experiences provide a vocabulary to describe examinee's experiences.

An example can help illustrate the different concepts and stages of the framework. Imagine you are going to take a test for college admission. You will prepare for this test (*pre-experience*), and you will interact with materials preparing you for the test and have different reactions to them during the preparation. The messages coming from the testing agency can affect your expectations about the test itself (dialogical level of the pre-experience) and what your friends and family think about the test (co-integrative level of the pre-experience).

When you take the test, you will experience many things in response to the test. You can correctly respond to an item and react as expected (cognitive interaction in the zero-experience on the free-stream level), or you can feel confused about the stem of an item (again, a cognitive interaction). You can respond well to an item but react to the stem (e.g., if you think it was racist in some way, which would be an expressive interaction), or you can be affected emotionally by the stem of an item if it speaks about something sensitive to you (emotional interaction). When you interact with a test, you may tend to first see a figure before reading the stem of a question (fluent interaction on the free-stream level), or you can have a strategy to respond to items in a certain way or a specific order (expressive interaction in the immediate level of the experience).

After you finish the test, you may have an opinion regarding the overall experience (expressive interaction on the integrative level) that you might share with your friends, which can change your perception of your test-taker experience (expressive interaction on the co-integrative level). That experience can lead you to participate in a rally against standardized tests because you feel that the test is racist (physical and expressive interaction on the dialogical level).

Figure 1 presents a summary of the three principles in the conceptual framework presented here about user experience.

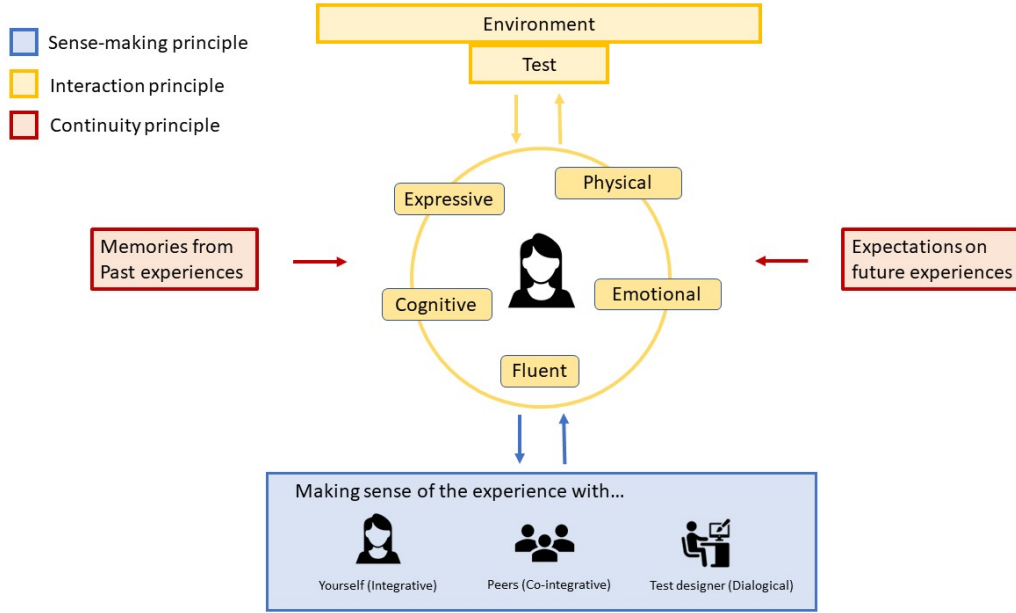


Figure 1: Diagram summarizing the Conceptual Framework for Test-Taker Experience.

1.4 Methods for UX Research in Educational Testing

The study of experience uses many varied methods. Qualitative methods such as think-aloud protocols are well-suited for the non-integrative levels of the experience. However, it is likely to be more difficult to measure other levels of experience. The GAMEX scale is a survey that tries to identify how gameful an experience was (Eppmann, Bekk, & Klein, 2018). Surveys can be an effective method for gathering information about test-takers' experiences.

Csikszentmihalyi and Larson (2014) use an "experience-sampling" survey methodology to capture information about a person's immediate and integrative experience during their daily routines. This method involves using a survey to gather experience data at random moments during the day. The survey includes open-ended questions and specific Likert-type questions about mood and the relevance of the activity a person was doing. Photovoice (Wang & Burris, 1997) is another method used in ethnography by which subjects take pictures and write a text about the picture to illustrate something about their experience and the topic the ethnographer wants to investigate.

It is essential to note that experience is not a single construct in the classical sense in which the word is typically used. A theoretical reduction of all experience to a set of measurable constructs will inevitably simplify and constrain the analysis of the experience. However, when we detect some problem related to the quality of the experience elicited by a test, we will want to create instruments that are particularly suited, like in the GAMEX scale. Thus, we believe that the study of user experiences in testing should be open-ended, to find and measure later specific constructs of relevance.

To study some aspects of experience, we may need to tell test-takers in advance to be particularly aware of certain occurrences such as their emotions or their cognitive difficulty when responding. This strategy requires persons aware of their experiences to help with experiential studies, like how sensitivity reviewers evaluate test items (Sireci & Mullane, 1994).

All analysis of direct observations of emotional states and reactions are sources of information about experience, bypassing the use of oral language. Similarly, brain scan imagery or other devices

that register information directly from the body can be used. For example, Dew, Galassi, and Galassi (1984) used physiological measures and questionnaires to study math test anxiety. Osborne (2007) used several physiological measures, such as skin conductance, heart rate, surface skin temperature, and blood pressure, to study gender stereotype threat on math proficiency. Maddox (2017) analyzed talk and gesture in the context of PIACC, tapping many types of interactions in the free stream, immediate, and dialogical levels of sense-making in a problem-solving task conducted on a computer and in the presence of an interviewer.

Another potentially valuable data source on examinee experience is log data from computer-based tests. These log data are often called “response process data” and can be part of the experience analysis if specific aspects of experience correspond to patterns in the log-data files (Araneda et al., 2022; Maddox, 2017; Goldhammer & Zehner, 2017). This tracing will require triangulation, linking some direct behavior registered in log-data files to an examinee’s experience. Questions asking about the immediate experience of test-takers in between items can be one way to triangulate log-data files and examinee experience.

All research related to response process can be considered part of UX research (Padilla & Benítez, 2014; Ercikan & Pellegrino, 2017; Zumbo & Hubley, 2017). Think-aloud protocols, cognitive labs, ethnographic studies, focus groups, and many other qualitative methods are a natural foundation for UX research in assessment.

Table 2 summarizes the different methods used to study examinee experience. Specific data collection methods apply for each combination of the level of sense-making and interaction type.

Other research methods relate to the temporal phases of a test-taker experience. Funds of knowledge (Moll et al., 1992) and socio-cognitive studies (Hubley et al., 2017) are theories and methods that help to understand the zero-experience in a contextualized way. Research on score reporting (e.g., Kannan et al., 2018; Zenisky & Hambleton, 2012) also includes a phenomenological dimension in the specific stage of receiving the results in a test-taker experience. Some researchers criticizing standardized tests and their consequences show aspects of the long-term experiences of test-takers (Walker-Johnson, 2009).

Table 2: Methods for Studying UX by Levels of Sense-making and Interaction Type

Level of Experience	Fluent	Cognitive	Expressive	Emotional	Physical
Free Stream	Direct observation, Log-data files	Think-aloud protocols, Log-data files	Think-aloud protocols	Direct observation of body/face, Think-aloud	Direct observation
Immediate	Direct observation, Log-data files	Cognitive interviews	Survey-like questions after each item	Survey-like questions after each item	Direct observation
Integrative	Log-data files, Interviews	Specific surveys	Surveys	Surveys	Surveys
Co-Integrative	Direct observation, Ethnographic studies	Focus groups, Surveys	Focus groups, Surveys	Focus groups, Surveys	Focus groups, Surveys
Dialogical	Direct observation, Ethnographic studies	Tailored focus groups, Surveys	Focus groups, Social media communication with agencies	Focus groups, Social media communication with agencies	Focus groups, Ethnographic studies

1.5 The Experiences of Test-takers Shared in Twitter: An Example from Chile

In the previous sections, we described how UX research methods could be applied to the study of test-takers’ experiences and presented a conceptual framework for doing so. Readers may wonder how practical it is to gather data on test-takers’ experiences. Although there are many methods (see Table 2), in this section, we present a short example of how to use the conceptual framework for TTX. Our example is in the context of a test used for college admission selection in Chile. We gathered information about tweets shared by test-takers and others about the high-stakes test used in Chile for university admission decisions, known as Prueba de Transición (PDT). This test happens during summer in the southern hemisphere (January), and it is the test battery used for university admission. In early 2021, the COVID-19 pandemic was spreading across the country as in other parts of the world, and there was a big question whether any large-scale standardized test, such as the PDT would be administered. The Chilean government and DEMRE – Departamento de Evaluación, Medición y Registro Educacional, the testing agency in charge of designing and administering the test – decided the test would be administered in person. Their decision included implementation of strict health precautions (e.g., the use of masks, proper ventilation of classrooms, etc.); however, students had to face a difficult decision of whether to take the test in person and risk exposure to the virus, or be safe at home and postpone their admission to universities one year.

2 Method

We connected to the Twitter API to collect tweets during the period May 2020 and March 2021, the period of time before and after the targeted instance of the test (taken in January 2021). We

looked for tweets to the DEMRE account on Twitter, or having the word “Prueba de Transición” on the tweets. We took a random sample from the total of tweets collected (12,000 approx.). After doing a manual filter for tweets that do not inform about test-taker experience, we came up with a subset of 516 tweets in total.

We recruited two raters to use the model for test-taker experience proposed in this paper, asking them to identify interactions on each tweet. Both raters were trained in a preliminary session, and a rubric was provided to use the assessment model in practice (see Appendix A).

After the two raters completed their identification task, Cohen’s Kappas were calculated for each type of interaction: Fluent, Cognitive, Expressive, Emotional and Physical. The two other levels in the test-taker experience were identified by the date of the tweets, or to whom it was directed; leaving only the interaction principle and categories to be determined by a rater.

We also calculated a correlation matrix (phi values) for the interaction types, using an OR operator to integrate raters (meaning that, we identify an interaction type present if at least one rater detected that type of interaction).

3 Results

Table 3 shows the Cohen’s Kappas for each of the interaction types included in our assessment model. Overall, all interaction types except Fluent interactions had fair to moderate agreement, according to the Landis and Koch (1977) criteria for interpreting Cohen’s kappa.

Table 3: Cohen’s Kappas for Raters Identifying Interactions Using our Proposed Model for Test-taker Experience

Type of Interaction	Cohen’s Kappa
Cognitive	0.61
Expressive	0.45
Emotional	0.40
Physical	0.33
Fluent	-0.02

We also calculated a matrix of Phi values for the nominal variables defined using an OR operator for both raters. The results are presented in Table 4.

Table 4: Correlation Matrix for OR-values of Detected Interactions Among Raters

	Cognitive	Expressive	Emotional	Physical	Fluent
Cognitive	1	-0.56	-0.49	-0.11	0.06
Expressive	-0.56	1	0.27	-0.02	-0.08
Emotional	-0.49	0.27	1	0.12	-0.08
Physical	-0.11	-0.02	0.12	1	-0.01
Fluent	0.06	-0.08	-0.08	-0.01	1

We observe negative correlations between cognitive and expressive, emotional, or physical interactions; positive correlation between expressive and emotional interactions; positive correlation between emotional and physical interactions; and the rest negligible correlations. The triad of

expressive–emotional–physical interaction has the potential to be one single factor; however, correlations are not high enough to support a strong conclusion. This finding contrasts with models for user experience that consider these constructs separately; however, those models have not produced empirical evidence for this claim.

Further research is needed to clarify this aspect of the model for test-taker experience. Nevertheless, the results of this study illustrate how aspects of test-taker experience can be gathered using social media to understand different stages of the testing experience.

4 Conclusion

In this paper, we discussed how UX methods can be broadly applied to educational testing to understand test-taker experiences. We also applied a method to analyze social media reports of experiences related to a high-stakes testing program in Chile. This case from Chile illustrates how UX research in testing can inform ways to enhance examinees’ experiences and generate hypotheses about potential sources of invalidity in the interpretations and uses of test results (Araneda, 2023). Further inquiry is required to gather evidence about the uses and interpretations of scores. For example, it would be valuable to examine how student opt-out decisions in their last year of high school were moderated by SES variables and by the distance to the testing site, above and beyond SES effects.

The example also illustrates how the conceptual framework proposed here can be used as a model for test-taker experience. First, the model allows the mapping of experience accounts within a three-dimensional space: (a) temporal stage in the test-taker experience, (b) level of sense-making, and (c) type of interaction predominant in the experience. This mapping can support systematic coding of narratives and other user-generated data about testing experiences.

In addition, the conceptual framework offers a broad notion of what constitutes user experience. Many efforts in UX research focus only on certain stages, interactions, or levels of sense-making, leaving many aspects unexplored. Our proposed framework provides criteria for identifying when we have conducted an exhaustive exploration of user experience across the three-dimensional space generated by Dewey’s principles of experience (Dewey, 1938).

One notable finding emerged regarding Fluent interactions. Although the literature mentions this category, raters in our study struggled to reach acceptable agreement. Upon discussion with the raters, we discovered differences in their interpretation of “fluent.” One rater did not believe subconscious interactions truly existed, while the other used the category as a catch-all. Although Forlizzi and Battarbee’s (2004) framework includes Fluency, it may be better conceptualized as an attribute of interactions rather than a separate category. For instance, riding a bicycle is a physical interaction that can be fluent—suggesting fluency should modify other categories, not stand alone. This finding indicates that our model may require refinement, with fluency being treated as a trait across interaction types. Further research is needed to clarify this aspect.

Educational testing requires a specific conceptualization of user experience that supports both enhancing the experience and validating the assessment. The conceptual framework proposed here can help unify the evidence traditionally known as response-process data with UX research, aligning with what Zumbo and Hubley (2017) envisioned as the future of response-process work. Dewey’s principles define the foundation of this framework:

- A *principle of interaction*, assuming that experience arises from a person’s engagement with their environment.
- A *principle of continuity*, assuming that all experience is shaped by past experiences and

expectations of future ones.

- A *principle of sense-making*, assuming that people interpret and re-interpret their experiences.

Our conceptual framework adopts these principles, defining categories aligned with each. User experience thus becomes a comprehensive source of validity evidence, extending traditional response-process research and potentially informing future editions of the *Standards* (AERA et al., 2014).

A phenomenological approach to testing allows us to redefine what we consider a “test.” A test becomes a device designed to elicit an experience from which we infer attributes about an examinee. Experience is what occurs within the interaction—not simply what is externally observable, and not the same for all users even under identical stimuli. The concept of UNDERSTANDARDization (Sireci, 2020) emphasizes that comparability should emerge from comparable experiences, not merely comparable stimuli.

A known risk in UX research is that researchers may misinterpret users’ experiences. This risk highlights the value of participatory approaches, ensuring interpretations align with what test-takers themselves would report. This risk exposes a limitation in the “test-taking experience” concept (Burstein et al., 2021): calling test-takers “test-takers” rather than “users” diminishes their agency, and the phrase itself underemphasizes all pre- and post-testing stages. The broader term *assessment experience* shifts attention toward understanding the full arc of experience.

Important insights arise when considering the dialogical level of experience. Scholars such as Markus and Borsboom (2013) and Hubley (2017) discuss testing as a conversational process. This perspective implies unexpected consequences: a poorly communicated validity argument may trigger negative reactions that introduce construct-irrelevant variance. A validity argument should therefore be rhetorically thoughtful and sensitive to the examinee’s perspective (Tindale, 2004). Dialogical missteps can become precursors of disengagement.

Test-takers are often unaware of the technical foundations of validity arguments, but messages communicated in those arguments can influence stakeholders and help construct shared narratives. Dialogical experience distinguishes between inferring about a machine and inferring about a human—humans respond to context. Initiatives like Randall’s (2021) anti-racist items illustrate this: beyond addressing fairness, such initiatives communicate care. Zumbo’s (2009, 2017) abductive reasoning approach to validity aligns with this view, emphasizing the caring intention behind explaining why a score was assigned.

We believe test developers should value the full spectrum of examinees’ experiences and consider the types of evidence that can be generated through UX research. User experience research is a powerful tool for identifying ethnocentric biases or narrow construct representations. In principle, test-taker data should be gathered throughout development and operational testing. While some validation efforts already focus on response-process evidence, our expansion to the entire experiential continuum deserves further exploration.

Although this framework arises from needs within educational testing, it can be extended to other educational technologies where understanding student–technology interactions is essential for design and validity. We hope the ideas presented here encourage deeper study of UX in testing and lead to improvements in educational and psychological assessments and technologies.

Use of AI Disclosure

We used different versions of GPT to improve the writing style, organization, and grammatical clarity of this manuscript. AI tools were also employed to assist with LaTeX formatting, table construction, and troubleshooting compilation issues.

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